

Name: _____ Date: _____

Period: _____

Planar Motion Check

A particle moves with $f(t) = (t^2 - 1, 2t - t^2)$ for $0 \leq t \leq 3$.

1. Complete the table.

t	$x(t) = t^2 - 1$	$y(t) = 2t - t^2$	(x, y)
0	_____	_____	_____
1	_____	_____	_____
2	_____	_____	_____
3	_____	_____	_____

2. Find the horizontal extrema of this motion. Show your work using the vertex formula for $x(t) = t^2 - 1$. Also check endpoints. _____

3. For what values of t does the particle cross the y -axis? Show your work, then state the coordinates of the particle at those moments. _____
